



Automatic generation of Slovenian traffic news for RTV Slovenija

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Abstract

Traffic reporting at RTV Slovenija is currently a manual process that relies on student workers compiling structured reports from the *promet.si* portal for bi-hourly broadcasts. This workflow is time-consuming and susceptible to human error. In this project, we investigate the potential of automating traffic news generation using large language models (LLMs). We explore several multilingual LLMs and evaluate their suitability for generating structured Slovenian traffic reports based on real-world data. Our methodology involves data preprocessing, prompt engineering techniques such as Role-Based and Few-Shot Prompting, and aligning data from structured traffic logs with their corresponding `.rtf` reports. A key challenge arises from inconsistencies between these two data sources, likely due to the use of additional, undocumented inputs in the original report generation. We describe our pipeline for linking and preprocessing the data, propose a framework for automatic and semi-automatic evaluation, and outline steps for future work focused on improving data alignment and testing advanced prompting strategies with the GaMS language model.

Keywords

traffic reports, natural language processing, large language models, prompt engineering

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Introduction

Traffic reporting is a crucial component of public broadcasting, providing real-time updates on road conditions, accidents, and congestion. At RTV Slovenija, traffic news is currently compiled manually by students who review traffic reports from the *promet.si* portal and transcribe them into structured news segments, which are then broadcast every 30 minutes. This manual approach is time-consuming, prone to human error, and inefficient given the volume of traffic data that needs to be processed.

With the advancement of large language models (LLMs), there is an opportunity to automate the generation of structured and readable traffic news. This project aims to leverage an existing LLM, apply prompt engineering techniques and fine-tune the model to automatically generate traffic reports that adhere to the broadcasting standards of RTV Slovenija. In the next chapter, we present various existing solutions that we consider using in this project.

Related Work and Existing Solutions

In their master's thesis, Bjørge and Lundegaard explored the automation of traffic incident reporting. They developed a system that collects real-time traffic data, processes it to identify incidents, and automatically generates reports for dissemination to the public and relevant authorities. Their work provides a detailed description of approaches that could be highly beneficial to our project [1].

LLM Selection

For this project, we consider three potential LLMs: *LLaMA*, *FLAN-T5*, and *Bloom*. These models have been pre-trained on multilingual datasets, including Slovenian text, enabling them to process Slovenian-language prompts. However, none of these models have been specifically trained on Slovenian traffic data, making fine-tuning essential for achieving optimal performance.

Both *LLaMA* and *FLAN-T5* demonstrate strong reasoning capabilities, making them well-suited for structured traffic report generation. In contrast, *Bloom* is less effective in reasoning tasks, which may lead to lower performance when applying *Chain-of-Thought Prompting*.

For evaluation, we think about combination of automatic and semi-automatic evaluation criteria. Automatic evaluation quantitatively assesses LLM outputs without human intervention. For this, we employ algorithms such as *BLEU*, which measures similarity to reference texts, and *ROUGE*, which evaluates recall-based overlap. However, for a more comprehensive assessment, human involvement is essential. Therefore, we incorporate semi-automatic evaluation, leveraging AI-assisted scoring. Specifically, we use another LLM, to grade generated outputs based on predefined criteria. Each score is then manually reviewed and adjusted as needed to ensure accuracy.

Prompt Engineering

To generate traffic reports, we consider using a combination of *Role-Based Prompting* and *Few-Shot Prompting*. This approach is particularly suitable because our dataset consists of structured traffic data paired with human-written traffic reports. By assigning the LLM the role of a traffic reporter and providing it with well-structured example reports, we aim to guide the model towards generating high-quality outputs.

Before report generation, the model will be given explicit instructions on how to structure the reports, which include:

- The location of the incident, specifying the affected road and direction.
- The cause of the disruption and its consequences.
- The specific section of the road impacted.

Additionally, the dataset contains road names that differ from those used in official reports, requiring the model to handle these discrepancies appropriately.

To ensure proper prioritization of events, an event hierarchy must be established, as some incidents are more critical than others. If the model fails to respect this hierarchy, we will experiment with *Chain-of-Thought Prompting* before fine-tuning the model. This technique introduces step-by-step reasoning, helping the model determine which incidents should be reported first.

Data Analysis

Reports Data From *promet.si*

The primary dataset is the `traffic_reports_2022_2023_2024.xlsx` file from the *promet.si* portal, which contains 170,823 traffic reports from the years 2022, 2023, and 2024, organized into 27 attributes, as illustrated in Figure 1, following with their descriptions. From these report data, LLM should be able to generate final report shown in the next subsection.

LegacyId Unique numerical identifier for the traffic report.

Datum Timestamp of when the traffic report was logged.

Operater Name and affiliation of the traffic report operator.

A1 Very important traffic alert message in Slovenian.

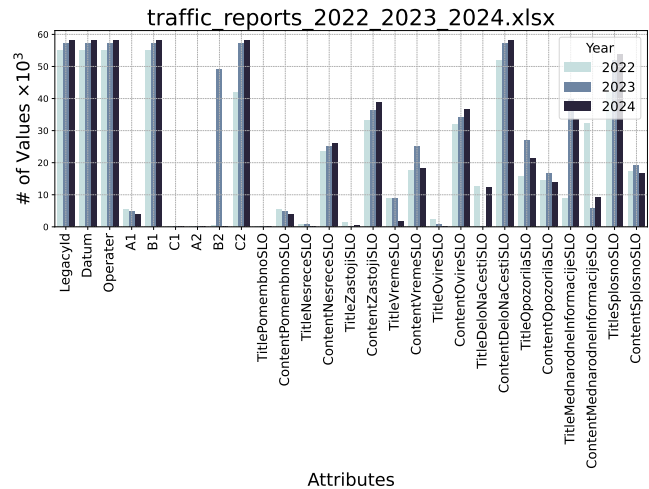


Figure 1. Project’s primary dataset visualization.

B1 Important traffic information in Slovenian.

C1 Less important information in Slovenian.

A2, B2, C2 English versions of A1, B1, C1.

Title+ContentPomembnoSLO Title and content of important Slovenian traffic information.

Title+ContentNesreceSLO Title and content related to road accidents.

Title+ContentZastojiSLO Title and content describing traffic congestion or delays.

Title+ContentVremeSLO Title and content related to weather conditions or weather-related traffic impacts.

Title+ContentOvireSLO Title and content concerning traffic obstacles (vehicle breakdowns, road hazards).

Title+ContentDeloNaCestiSLO Title and content related to roadworks or construction affecting traffic.

Title+ContentOpozorilaSLO Title and content containing public warnings, announcements, or safety reminders.

Title+ContentMednarodneInformacijeSLO Title and content regarding international or cross-border traffic information.

Title+ContentSplosnoSLO Title and content of general traffic information not covered by other specific categories.

Final Reports From RTV Slovenija

The second important dataset consists of 28,037 `.rtf` files, each containing desired output reports from RTV Slovenija, collected monthly throughout the years 2022, 2023, and 2024. An example of such a report is shown below:

NOVE prometne informacije 31. 01. 2022
18:30 1. in 3. program

Podatki o prometu.

Zaradi zdrsra tovornih vozil promet poteka izmenično enosmerno na glavni cesti Novo mesto – Metlika pri Vahti ter na regionalni cesti Soteska – Črmošnjice – Črnomelj pri Starih Žagah.

Na mejnih prehodih Gruškovje in Obrežje vozniki tovornih vozil na vstop v državo čakajo eno uro. Na hrvaški strani mejnega prehoda Babno Polje je prepovedan promet za priklopnike in polpriklopnike.

Ponekod na Dolenjskem sneži. Predvsem na območju med Novim mestom, Kočevjem in Črnomljem se sneg oprijema vozišča. Tudi drugod po državi so ceste mokre in spolzke, lahko tudi poledenele. Vozite previdno, prilagodite hitrost in povečajte varnostno razdaljo.

In addition, we also received guidelines from RTV Slovenija on how the reports should be properly structured — including conventions on road naming, sentence ordering by priority, and other formatting details.

Data Preprocessing

Due to a substantial amount of missing or irrelevant data, we reduced the number of attributes to those highlighted in [blue](#). Additionally, we introduced a new attribute, [RTF file name](#), which enables linking each traffic record to its corresponding `.rtf` file based on the date of the report.

The dataset from *promet.si* required further preprocessing, as many of its values contained embedded HTML content. Similarly, the reports extracted from `.rtf` files had to be converted into a structured DataFrame format in order to facilitate linking with traffic data and to enable automated prompt generation.

For subsequent model training and evaluation, we partitioned the data into training (70%), testing (15%), and validation (15%) sets. Since each entry in traffic data from *promet.si* corresponds to a specific `.rtf` file, only the `.rtf` files needed to be split accordingly.

Methods

Few-shot Prompting

For few-shot prompting, we constructed prompts by combining relevant data into an example input for one test sample. As shown on example below this section, the information extracted from the structured traffic data and the content of the corresponding `.rtf` files **do not align**.

To verify the cause of this mismatch, we manually cross-checked several rows in the `.xlsx` file with their corresponding `.rtf` files using the report dates as a reference. Based on this inspection, we concluded that the issue does not stem from an error in our data-linking process, but rather from

inconsistencies in the provided datasets. **We assume that additional, untracked data sources were used in the creation of the this `.rtf` reports.**

An example of a constructed part of few-shot input that informations do not match:

Vhodni podatki:

- **Manj pomembne informacije o prometu:**

Na avtocesti od Sežane proti Mariboru, do Slovenskih Konjic, pelje izredni prevoz. Občasno je lahko promet oviran in upočasnjen. Čakalna doba je na mejnem prehodu Obrežje. Zaradi praznikov bo po Sloveniji v nedeljo, 1. 5., in v ponedeljek, 2. 5., med 8. in 22. uro, veljala omejitev prometa tovornih vozil, katerih največja dovoljena masa presega 7,5 t.

- **Informacije o opozorilih:** Na avtocesti od Sežane proti Mariboru, do Slovenskih Konjic, pelje izredni prevoz. Občasno je lahko promet oviran in upočasnjen.

- **Informacije o mednarodnih informacijah:** Čakalna doba je na mejnem prehodu Obrežje.

- **Splošne informacije:** Zaradi praznikov bo po Sloveniji v nedeljo, 1. 5., in v ponedeljek, 2. 5., med 8. in 22. uro, veljala omejitev prometa tovornih vozil, katerih največja dovoljena masa presega 7,5 t.

Poročilo:

Prometne informacije – 30. 04. 2022, 18:30, 1. in 2. program.

Zaradi prometne nesreče je zaprta regionalna cesta Ajševica–Rožna Dolina, in to pri Ajševici. Na mejnem prehodu Obrežje vozniki na vstop v državo čakajo do dve uri, v Gruškovju pa pol ure. Povečan promet pri izstopu iz države pa je na prehodu Dobovec, na katerem vozniki čakajo uro in pol, ter na Obrežju in v Gruškovju, v katerem vozniki čakajo pol ure.

For future work, the first priority is to align and clean the data to ensure that prompt engineering with the GaMS large language model is feasible. Once this is achieved, we plan to experiment with and evaluate various prompt engineering strategies described in the paragraph Related Work and Existing Solutions, time permitting.

References

- [1] Anna Jansdatter Bjørge and Mads Lundegaard. Automating the process of traffic incident reporting. Master's thesis, NTNU, 2024.